13

Concurrency

Keep it simple:
as simple as possible,
but no simpler.
– A. Einstein

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13.1 Introduction

Concurrency – the execution of several tasks simultaneously – is widely used to improve throughput (by using several processors for a single computation) or to improve responsiveness (by allowing one part of a program to progress while another is waiting for a response). All modern programming languages provide support for this. The support provided by the C++ standard library is a portable and type-safe variant of what has been used in C++ for more than 20 years and is almost universally supported by modern hardware. The standard-library support is primarily aimed at supporting systems-level concurrency rather than directly providing sophisticated higher-level concurrency models; those can be supplied as libraries built using the standard-library facilities.

The standard library directly supports concurrent execution of multiple threads in a single address space. To allow that, C++ provides a suitable memory model and a set of atomic operations. The atomic operations allows lock-free programming [Dechev,2012]. The memory model

ensures that as long as a programmer avoids data races (uncontrolled concurrent access to mutable data), everything works as one would naively expect. However, most users will see concurrency only in terms of the standard library and libraries built on top of that. This section briefly gives examples of the main standard-library concurrency support facilities: **thread**s, **mutex**es, **lock()** operations, **packaged_task**s, and **future**s. These features are built directly upon what operating systems offer and do not incur performance penalties compared with those. Neither do they guarantee significant performance improvements compared to what the operating system offers.

Do not consider concurrency a panacea. If a task can be done sequentially, it is often simpler and faster to do so.

13.2 Tasks and threads

We call a computation that can potentially be executed concurrently with other computations a *task*. A *thread* is the system-level representation of a task in a program. A task to be executed concurrently with other tasks is launched by constructing a **std::thread** (found in **<thread>**) with the task as its argument. A task is a function or a function object:

```
void f();                   // function

struct F {                  // function object
    void operator()();      // F's call operator (§5.5)
};

void user()
{
    thread t1 {f};          // f() executes in separate thread
    thread t2 {F()};        // F()() executes in separate thread

    t1.join();              // wait for t1
    t2.join();              // wait for t2
}
```

The **join()**s ensure that we don't exit **user()** until the threads have completed. To "join" a **thread** means to "wait for the thread to terminate."

Threads of a program share a single address space. In this, threads differ from processes, which generally do not directly share data. Since threads share an address space, they can communicate through shared objects (§13.5). Such communication is typically controlled by locks or other mechanisms to prevent data races (uncontrolled concurrent access to a variable).

Programming concurrent tasks can be *very* tricky. Consider possible implementations of the tasks **f** (a function) and **F** (a function object):

```
void f() { cout << "Hello "; }

struct F {
    void operator()() { cout << "Parallel World!\n"; }
};
```

This is an example of a bad error: Here, **f** and **F()** each use the object **cout** without any form of

synchronization. The resulting output would be unpredictable and could vary between different executions of the program because the order of execution of the individual operations in the two tasks is not defined. The program may produce "odd" output, such as

```
PaHerallllel o World!
```

When defining tasks of a concurrent program, our aim is to keep tasks completely separate except where they communicate in simple and obvious ways. The simplest way of thinking of a concurrent task is as a function that happens to run concurrently with its caller. For that to work, we just have to pass arguments, get a result back, and make sure that there is no use of shared data in between (no data races).

13.3 Passing Arguments

Typically, a task needs data to work upon. We can easily pass data (or pointers or references to the data) as arguments. Consider:

```
void f(vector<double>& v);      // function do something with v

struct F {                      // function object: do something with v
        vector<double>& v;
        F(vector<double>& vv) :v{vv} { }
        void operator()();      // application operator; §5.5
};

int main()
{
        vector<double> some_vec {1,2,3,4,5,6,7,8,9};
        vector<double> vec2 {10,11,12,13,14};

        thread t1 {f,ref(some_vec)};    // f(some_vec) executes in a separate thread
        thread t2 {F{vec2}};            // F(vec2)() executes in a separate thread

        t1.join();
        t2.join();
}
```

Obviously, `F{vec2}` saves a reference to the argument vector in `F`. `F` can now use that vector and hopefully no other task accesses `vec2` while `F` is executing. Passing `vec2` by value would eliminate that risk.

The initialization with `{f,ref(some_vec)}` uses a `thread` variadic template constructor that can accept an arbitrary sequence of arguments (§5.6). The `ref()` is a type function from `<functional>` that unfortunately is needed to tell the variadic template to treat `some_vec` as a reference, rather than as an object. The compiler checks that the first argument can be invoked given the following arguments and builds the necessary function object to pass to the thread. Thus, if `F::operator()()` and `f()` perform the same algorithm, the handling of the two tasks are roughly equivalent: in both cases, a function object is constructed for the `thread` to execute.

13.4 Returning Results

In the example in §13.3, I pass the arguments by non-**const** reference. I only do that if I expect the task to modify the value of the data referred to (§1.8). That's a somewhat sneaky, but not uncommon, way of returning a result. A less obscure technique is to pass the input data by **const** reference and to pass the location of a place to deposit the result as a separate argument:

```
void f(const vector<double>& v, double* res);      // take input from v;  place result in *res

class F {
public:
      F(const vector<double>& vv, double* p) :v{vv}, res{p} { }
      void operator()();                 // place result in *res
private:
      const vector<double>& v;           // source of input
      double* res;                       // target for output
};

int main()
{
      vector<double> some_vec;
      vector<double> vec2;
      // ...

      double res1;
      double res2;

      thread t1 {f,cref(some_vec),&res1};      // f(some_vec,&res1) executes in a separate thread
      thread t2 {F{vec2,&res2}};               // F{vec2,&res2}() executes in a separate thread

      t1.join();
      t2.join();

      cout << res1 << ' ' << res2 << '\n';
}
```

This works and the technique is very common, but I don't consider returning results through arguments particularly elegant, so I return to this topic in §13.7.1.

13.5 Sharing Data

Sometimes tasks need to share data. In that case, the access has to be synchronized so that at most one task at a time has access. Experienced programmers will recognize this as a simplification (e.g., there is no problem with many tasks simultaneously reading immutable data), but consider how to ensure that at most one task at a time has access to a given set of objects.

The fundamental element of the solution is a **mutex**, a "mutual exclusion object." A **thread** acquires a mutex using a **lock()** operation:

```
mutex m; // controlling mutex
int sh;     // shared data

void f()
{
        unique_lock<mutex> lck {m};  // acquire mutex
        sh += 7;                     // manipulate shared data
}       // release mutex implicitly
```

The **unique_lock**'s constructor acquires the mutex (through a call **m.lock()**). If another thread has already acquired the mutex, the thread waits (''blocks'') until the other thread completes its access. Once a thread has completed its access to the shared data, the **unique_lock** releases the **mutex** (with a call **m.unlock()**). When a **mutex** is released, **thread**s waiting for it resume executing (''are woken up''). The mutual exclusion and locking facilities are found in **<mutex>**.

The correspondence between the shared data and a **mutex** is conventional: the programmer simply has to know which **mutex** is supposed to correspond to which data. Obviously, this is error-prone, and equally obviously we try to make the correspondence clear through various language means. For example:

```
class Record {
public:
        mutex rm;
        // ...
};
```

It doesn't take a genius to guess that for a **Record** called **rec**, **rec.rm** is a **mutex** that you are supposed to acquire before accessing the other data of **rec**, though a comment or a better name might have helped a reader.

It is not uncommon to need to simultaneously access several resources to perform some action. This can lead to deadlock. For example, if **thread1** acquires **mutex1** and then tries to acquire **mutex2** while **thread2** acquires **mutex2** and then tries to acquire **mutex1**, then neither task will ever proceed further. The standard library offers help in the form of an operation for acquiring several locks simultaneously:

```
void f()
{
        // ...
        unique_lock<mutex> lck1 {m1,defer_lock};    // defer_lock: don't yet try to acquire the mutex
        unique_lock<mutex> lck2 {m2,defer_lock};
        unique_lock<mutex> lck3 {m3,defer_lock};
        // ...
        lock(lck1,lck2,lck3);                       // acquire all three locks
        // ... manipulate shared data ...
} // implicitly release all mutexes
```

This **lock()** will proceed only after acquiring all its **mutex** arguments and will never block (''go to sleep'') while holding a **mutex**. The destructors for the individual **unique_lock**s ensure that the **mutex**es are released when a **thread** leaves the scope.

Communicating through shared data is pretty low level. In particular, the programmer has to devise ways of knowing what work has and has not been done by various tasks. In that regard, use of shared data is inferior to the notion of call and return. On the other hand, some people are convinced that sharing must be more efficient than copying arguments and returns. That can indeed be so when large amounts of data are involved, but locking and unlocking are relatively expensive operations. On the other hand, modern machines are very good at copying data, especially compact data, such as **vector** elements. So don't choose shared data for communication because of "efficiency" without thought and preferably not without measurement.

13.6 Waiting for Events

Sometimes, a **thread** needs to wait for some kind of external event, such as another **thread** completing a task or a certain amount of time having passed. The simplest "event" is simply time passing. Using the time facilities found in **<chrono>** I can write:

```
using namespace std::chrono;      // see §11.4

auto t0 = high_resolution_clock::now();
this_thread::sleep_for(milliseconds{20});
auto t1 = high_resolution_clock::now();

cout << duration_cast<nanoseconds>(t1–t0).count() << " nanoseconds passed\n";
```

Note that I didn't even have to launch a **thread**; by default, **this_thread** refers to the one and only thread.

I used **duration_cast** to adjust the clock's units to the nanoseconds I wanted.

The basic support for communicating using external events is provided by **condition_variable**s found in **<condition_variable>**. A **condition_variable** is a mechanism allowing one **thread** to wait for another. In particular, it allows a **thread** to wait for some *condition* (often called an *event*) to occur as the result of work done by other **thread**s.

Using **condition_variable**s supports many forms of elegant and efficient sharing, but can be rather tricky. Consider the classical example of two **thread**s communicating by passing messages through a **queue**. For simplicity, I declare the **queue** and the mechanism for avoiding race conditions on that **queue** global to the producer and consumer:

```
class Message {      // object to be communicated
    // ...
};

queue<Message> mqueue;          // the queue of messages
condition_variable mcond;       // the variable communicating events
mutex mmutex;                   // the locking mechanism
```

The types **queue**, **condition_variable**, and **mutex** are provided by the standard library.

The **consumer()** reads and processes **Message**s:

```
void consumer()
{
    while(true) {
        unique_lock<mutex> lck{mmutex};        // acquire mmutex
        while (mcond.wait(lck)) /* do nothing */;  // release lck and wait;
                                               // re-acquire lck upon wakeup
        auto m = mqueue.front();               // get the message
        mqueue.pop();
        lck.unlock();                          // release lck
        // ... process m ...
    }
}
```

Here, I explicitly protect the operations on the **queue** and on the **condition_variable** with a **unique_lock** on the **mutex**. Waiting on **condition_variable** releases its lock argument until the wait is over (so that the queue is non-empty) and then reacquires it.

The corresponding **producer** looks like this:

```
void producer()
{
    while(true) {
        Message m;
        // ... fill the message ...
        unique_lock<mutex> lck {mmutex};       // protect operations
        mqueue.push(m);
        mcond.notify_one();                    // notify
    }                                          // release lock (at end of scope)
}
```

13.7 Communicating Tasks

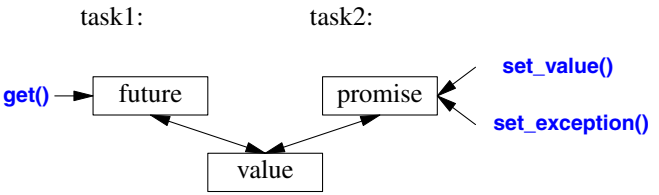
The standard library provides a few facilities to allow programmers to operate at the conceptual level of tasks (work to potentially be done concurrently) rather than directly at the lower level of threads and locks:

[1] **future** and **promise** for returning a value from a task spawned on a separate thread
[2] **packaged_task** to help launch tasks and connect up the mechanisms for returning a result
[3] **async()** for launching of a task in a manner very similar to calling a function.

These facilities are found in **<future>**.

13.7.1 future and promise

The important point about **future** and **promise** is that they enable a transfer of a value between two tasks without explicit use of a lock; "the system" implements the transfer efficiently. The basic idea is simple: When a task wants to pass a value to another, it puts the value into a **promise**. Somehow, the implementation makes that value appear in the corresponding **future**, from which it can be read (typically by the launcher of the task). We can represent this graphically:



If we have a future<X> called fx, we can get() a value of type X from it:

```
X v = fx.get();   // if necessary, wait for the value to get computed
```

If the value isn't there yet, our thread is blocked until it arrives. If the value couldn't be computed, get() might throw an exception (from the system or transmitted from the task from which we were trying to get() the value).

The main purpose of a promise is to provide simple "put" operations (called set_value() and set_exception()) to match future's get(). The names "future" and "promise" are historical; please don't blame or credit me. They are yet another fertile source of puns.

If you have a promise and need to send a result of type X to a future, you can do one of two things: pass a value or pass an exception. For example:

```
void f(promise<X>& px)  // a task: place the result in px
{
        // ...
        try {
                X res;
                // ... compute a value for res ...
                px.set_value(res);
        }
        catch (...) {           // oops: couldn't compute res
                px.set_exception(current_exception());          // pass the exception to the future's thread
        }
}
```

The current_exception() refers to the caught exception.

To deal with an exception transmitted through a future, the caller of get() must be prepared to catch it somewhere. For example:

```
void g(future<X>& fx)           // a task: get the result from fx
{
        // ...
        try {
                X v = fx.get();  // if necessary, wait for the value to get computed
                // ... use v ...
        }
        catch (...) {           // oops: someone couldn't compute v
                // ... handle error ...
        }
}
```

If the error doesn't need to be handled by **g()** itself, the code reduces to the minimal:

```
void g(future<X>& fx)          // a task: get the result from fx
{
    // ...
    X v = fx.get();   // if necessary, wait for the value to get computed
    // ... use v ...
}
```

13.7.2 packaged_task

How do we get a **future** into the task that needs a result and the corresponding **promise** into the thread that should produce that result? The **packaged_task** type is provided to simplify setting up tasks connected with **future**s and **promise**s to be run on **thread**s. A **packaged_task** provides wrapper code to put the return value or exception from the task into a **promise** (like the code shown in §13.7.1). If you ask it by calling **get_future**, a **packaged_task** will give you the **future** corresponding to its **promise**. For example, we can set up two tasks to each add half of the elements of a **vector<double>** using the standard-library **accumulate()** (§12.3):

```
double accum(double* beg, double* end, double init)
    // compute the sum of [beg:end) starting with the initial value init
{
    return accumulate(beg,end,init);
}

double comp2(vector<double>& v)
{
    using Task_type = double(double*,double*,double);          // type of task

    packaged_task<Task_type> pt0 {accum};                      // package the task (i.e., accum)
    packaged_task<Task_type> pt1 {accum};

    future<double> f0 {pt0.get_future()};                      // get hold of pt0's future
    future<double> f1 {pt1.get_future()};                      // get hold of pt1's future

    double* first = &v[0];
    thread t1 {move(pt0),first,first+v.size()/2,0};            // start a thread for pt0
    thread t2 {move(pt1),first+v.size()/2,first+v.size(),0};   // start a thread for pt1

    // ...

    return f0.get()+f1.get();                                  // get the results
}
```

The **packaged_task** template takes the type of the task as its template argument (here **Task_type**, an alias for **double(double*,double*,double)**) and the task as its constructor argument (here, **accum**). The **move()** operations are needed because a **packaged_task** cannot be copied. The reason that a **packaged_task** cannot be copied is that it is a resource handle: it owns its **promise** and is (indirectly) responsible for whatever resoures its task may own.

Please note the absence of explicit mention of locks in this code: we are able to concentrate on tasks to be done, rather than on the mechanisms used to manage their communication. The two tasks will be run on separate threads and thus potentially in parallel.

13.7.3 async()

The line of thinking I have pursued in this chapter is the one I believe to be the simplest yet still among the most powerful: Treat a task as a function that may happen to run concurrently with other tasks. It is far from the only model supported by the C++ standard library, but it serves well for a wide range of needs. More subtle and tricky models, e.g., styles of programming relying on shared memory, can be used as needed.

To launch tasks to potentially run asynchronously, we can use **async()**:

```
double comp4(vector<double>& v)
    // spawn many tasks if v is large enough
{
    if (v.size()<10000)          // is it worth using concurrency?
        return accum(v.begin(),v.end(),0.0);

    auto v0 = &v[0];
    auto sz = v.size();

    auto f0 = async(accum,v0,v0+sz/4,0.0);          // first quarter
    auto f1 = async(accum,v0+sz/4,v0+sz/2,0.0);     // second quarter
    auto f2 = async(accum,v0+sz/2,v0+sz*3/4,0.0);   // third quarter
    auto f3 = async(accum,v0+sz*3/4,v0+sz,0.0);     // fourth quarter

    return f0.get()+f1.get()+f2.get()+f3.get();  // collect and combine the results
}
```

Basically, **async()** separates the "call part" of a function call from the "get the result part," and separates both from the actual execution of the task. Using **async()**, you don't have to think about threads and locks. Instead, you think just in terms of tasks that potentially compute their results asynchronously. There is an obvious limitation: Don't even think of using **async()** for tasks that share resources needing locking – with **async()** you don't even know how many **thread**s will be used because that's up to **async()** to decide based on what it knows about the system resources available at the time of a call. For example, **async()** may check whether any idle cores (processors) are available before deciding how many **thread**s to use.

Using a guess about the cost of computation relative to the cost of launching a **thread**, such as **v.size()<10000**, is very primitive and prone to gross mistakes about performance. However, this is not the place for a proper disussion about how to manage **thread**s. Don't take this estimate as more than a simple and probably poor guess.

Please note that **async()** is not just a mechanism specialized for parallel computation for increased performance. For example, it can also be used to spawn a task for getting information from a user, leaving the "main program" active with something else (§13.7.3).

13.8 Advice

[1] The material in this chapter roughly corresponds to what is described in much greater detail in Chapters 41-42 of [Stroustrup,2013].
[2] Use concurrency to improve responsiveness or to improve throughput; §13.1.
[3] Work at the highest level of abstraction that you can afford; §13.1.
[4] Consider processes as an alternative to threads; §13.1.
[5] The standard-library concurrency facilities are type safe; §13.1.
[6] The memory model exists to save most programmers from having to think about the machine architecture level of computers; §13.1.
[7] The memory model makes memory appear roughly as naively expected; §13.1.
[8] Atomics allow for lock-free programming; §13.1.
[9] Leave lock-free programming to experts; §13.1.
[10] Sometimes, a sequential solution is simpler and faster than a concurrent solution; §13.1.
[11] Avoid data races; §13.1, §13.2.
[12] A **thread** is a type-safe interface to a system thread; §13.2.
[13] Use **join()** to wait for a **thread** to complete; §13.2.
[14] Avoid explicitly shared data whenever you can; §13.2.
[15] Use **unique_lock** to manage mutexes; §13.5.
[16] Use **lock()** to acquire multiple locks; §13.5.
[17] Use **condition_variable**s to manage communication among **thread**s; §13.6.
[18] Think in terms of tasks that can be executed concurrently, rather than directly in terms of **thread**s; §13.7.
[19] Value simplicity; §13.7.
[20] Prefer **packaged_task** and **future**s over direct use of **thread**s and **mutex**es; §13.7.
[21] Return a result using a **promise** and get a result from a **future**; §13.7.1.
[22] Use **packaged_task**s to handle exceptions thrown by tasks and to arrange for value return; §13.7.2.
[23] Use a **packaged_task** and a **future** to express a request to an external service and wait for its response; §13.7.2.
[24] Use **async()** to launch simple tasks; §13.7.3.